GROUP PROJECT

December 10

Course

Digital Logic Design

SCENARIO

Scenario 2
Access Control & Indicators

Scenario 3
Mini Vending Controller

MINI VENDING CONTROLLER

Inputs

- C1 (1-unit coin)
- C2 (2-unit coin)
- S (Select)
- O (Out of stock)

Outputs

- Dispense
- ReturnCoin
- ThankLED
- Change
- Error
- ServiceLED

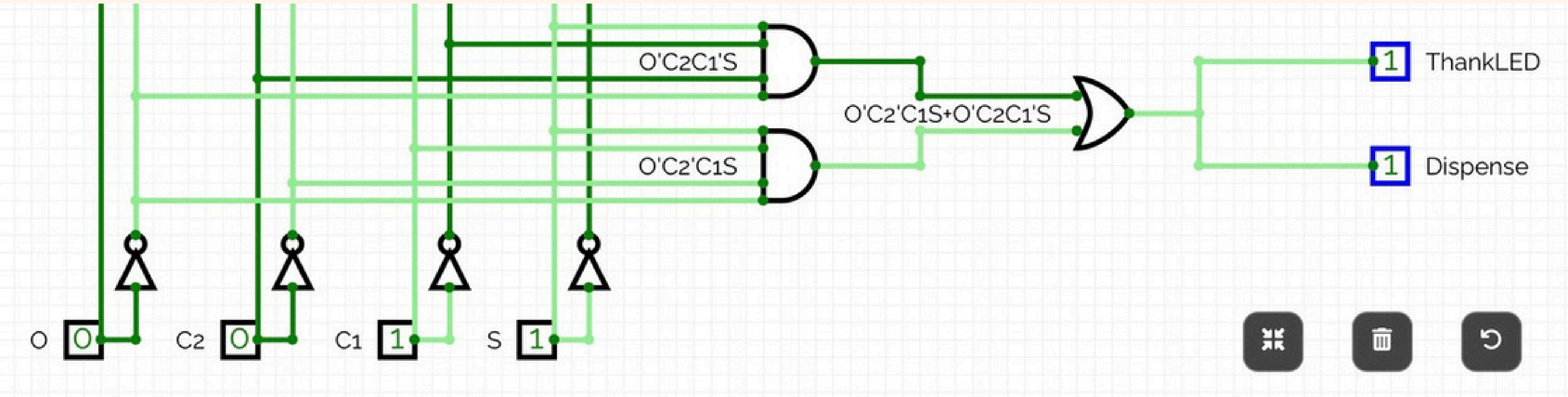
MINI VENDING CONTROLLER

- Dispense

0	0	1	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: Σ (3,5)

SOP= O' C2' C1 S + O' C2 C1' S

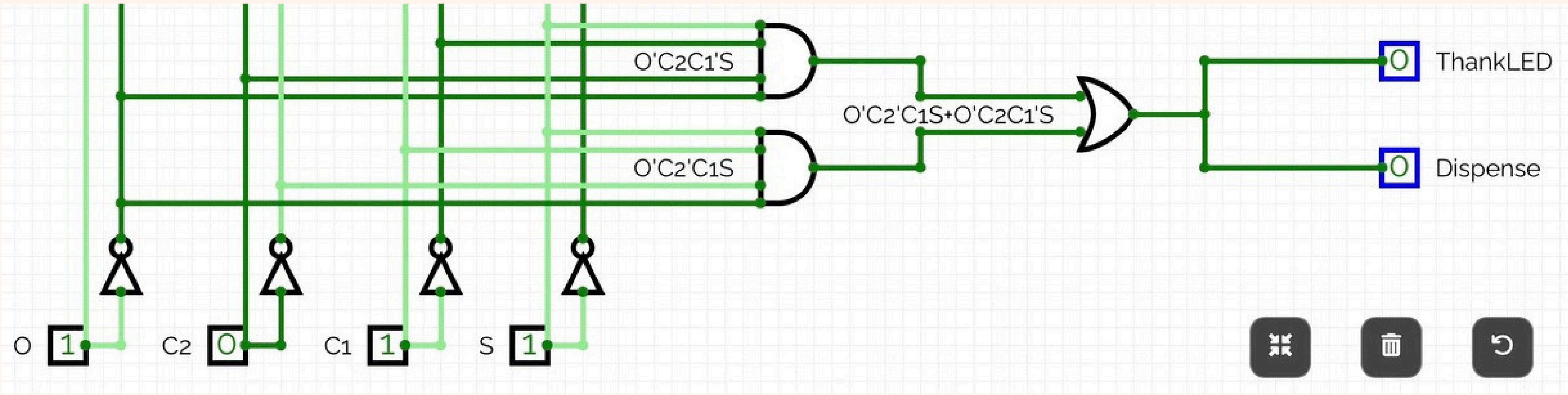


- ThankLED :

0	0	1	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: Σ (3,5)

SOP = O' C2' C1 S + O' C2 C1' S



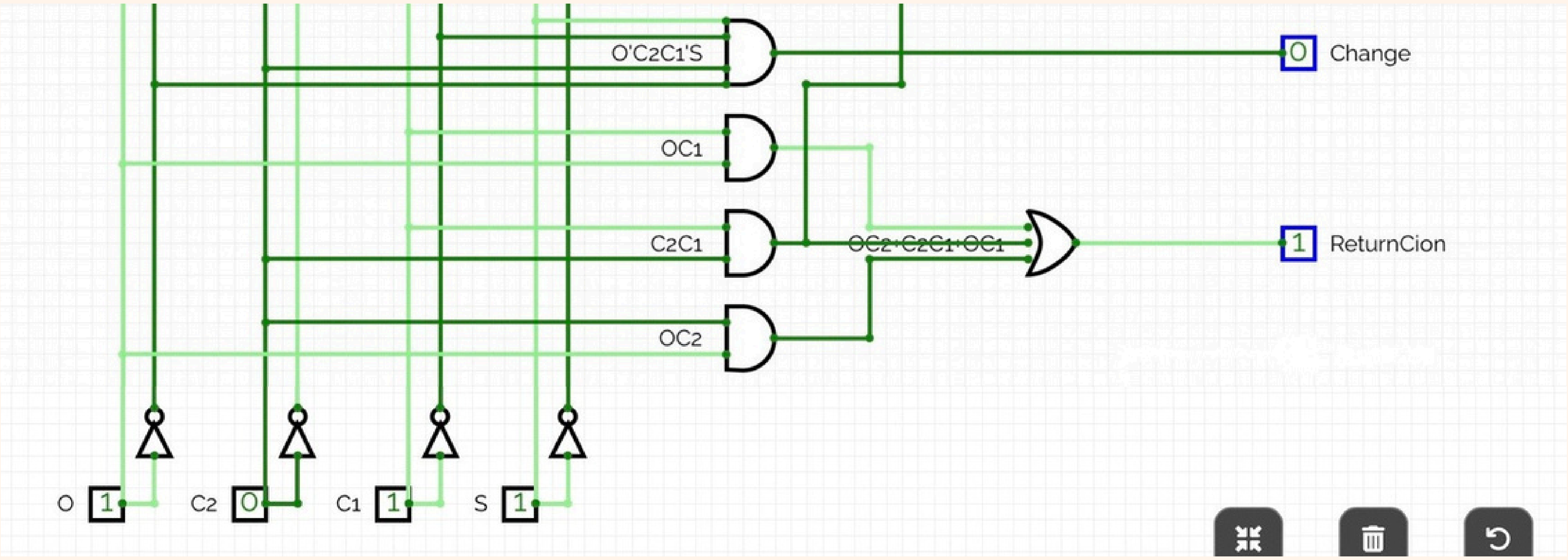
MINI VENDING CONTROLLER

- ReturnCoin

0	0	0	0
0	1	1	1
1	1	1	1
0	0	1	1

Minterms: Σ (6,7,10,11,12,13,14,15)

SOP =O C2 + C2 C1 + O C2

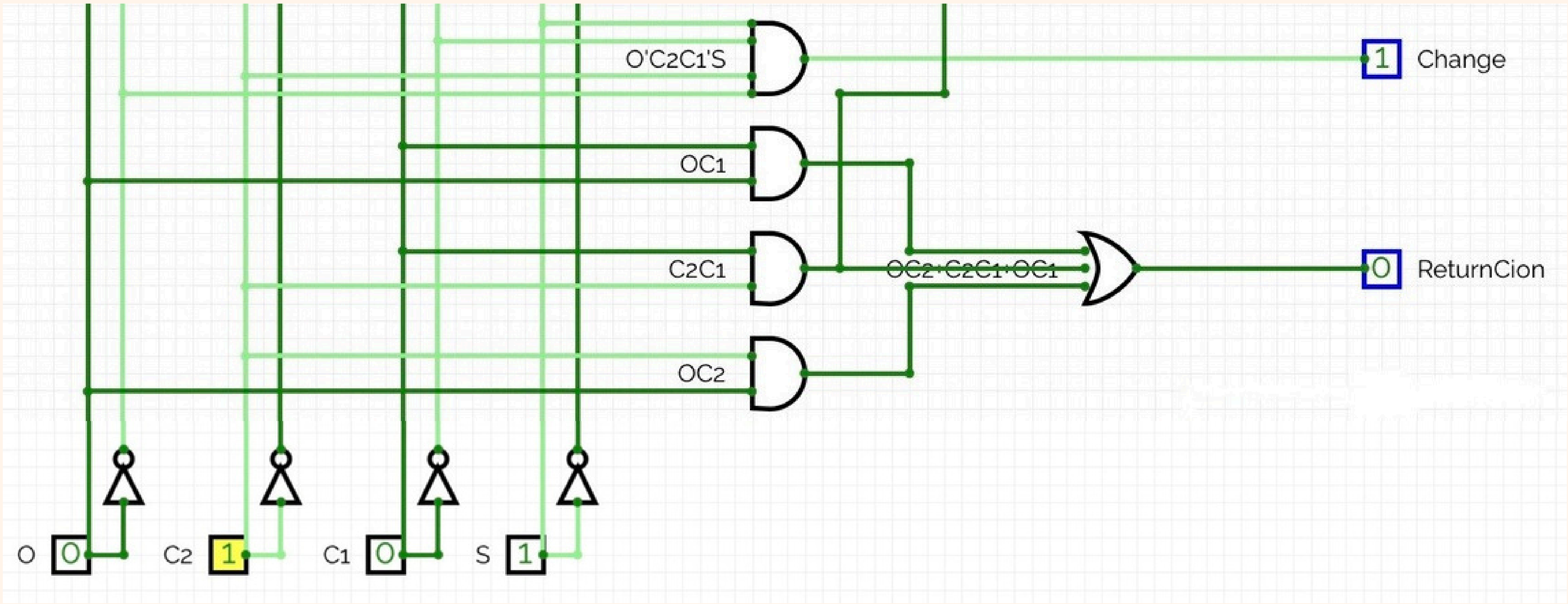


- Change :

0	0	0	0
0	1	0	0
0	0	0	0
0	0	0	0

Minterms: Σ (5)

SOP =O' C2 C1'



MINI VENDING CONTROLLER

- Error:

0	1	1	0
0	1	1	1
1	1	1	1
1	1	1	1

Minterms: Σ (1,6,7,8,9,10,11,12,13,14,15)

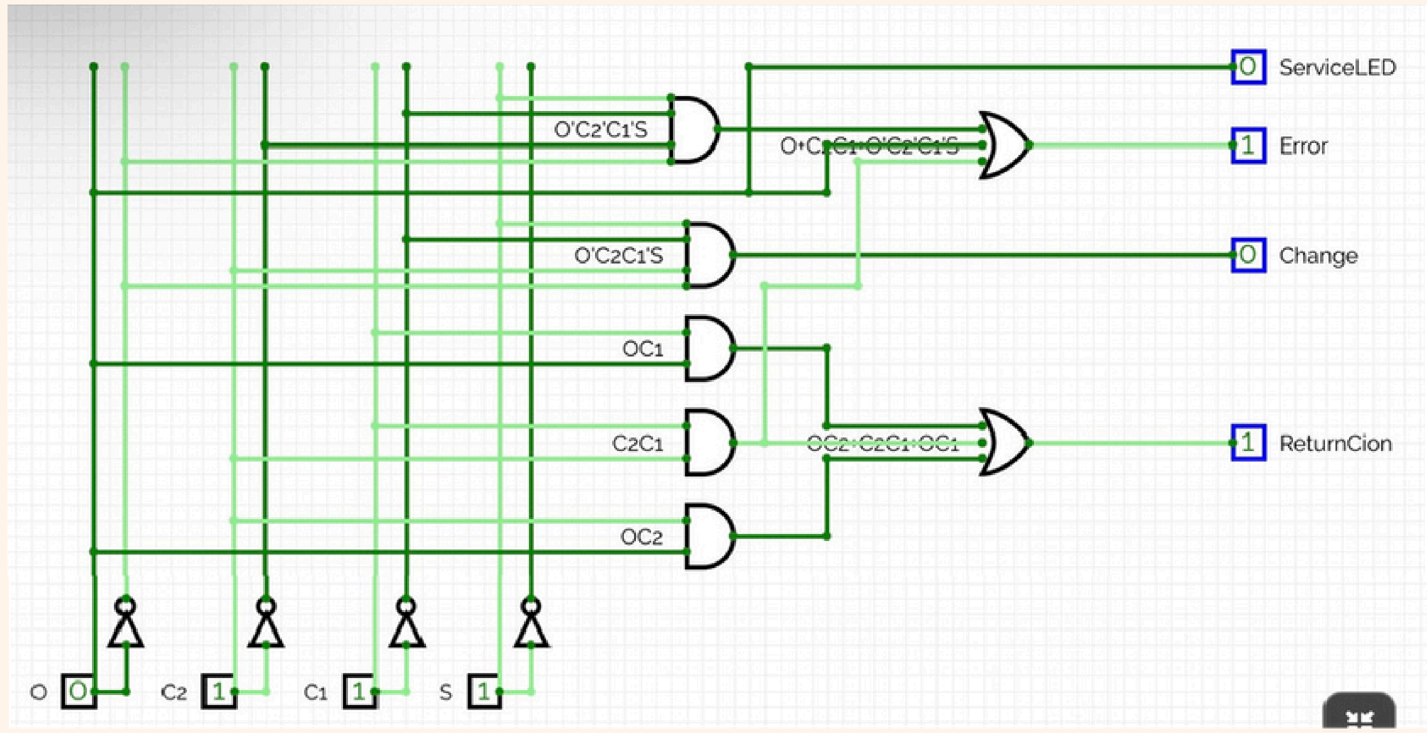
SOP = O + C2 C1 + O' C2' C1' S

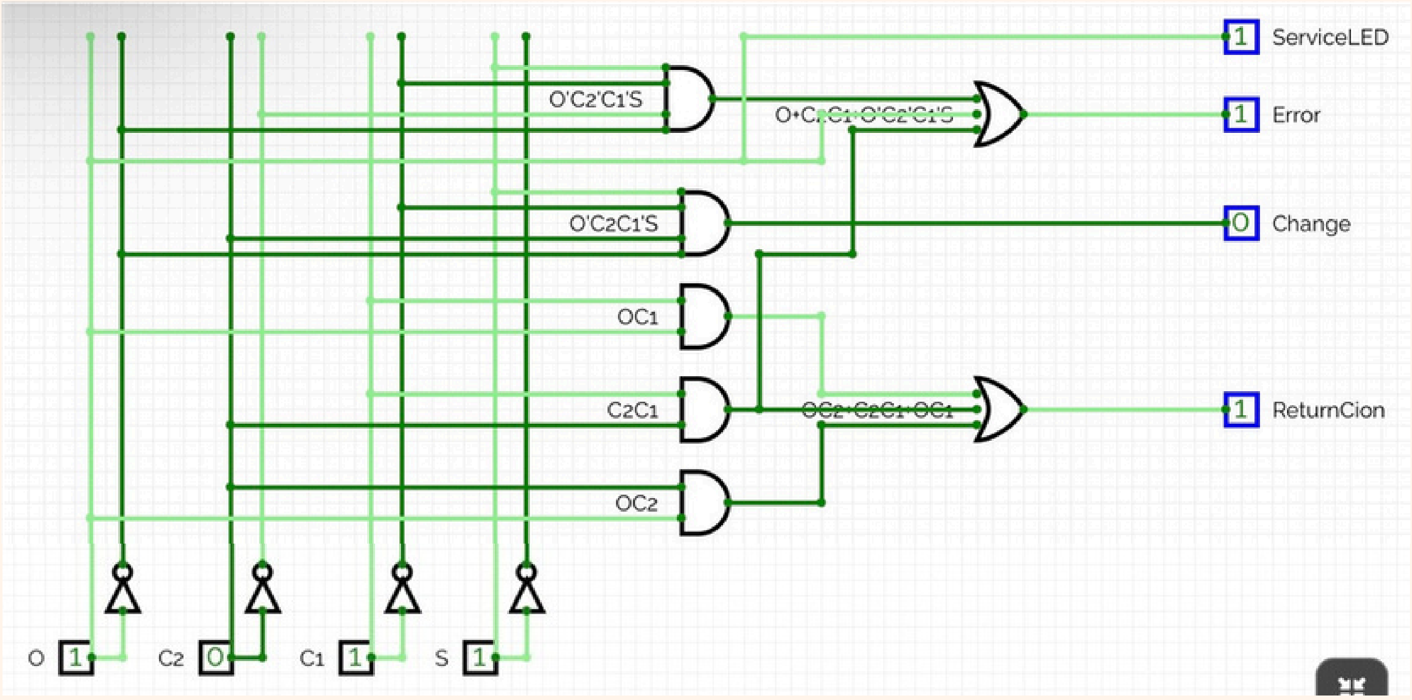
- SerriceLED :

0	0	0	0
0	1	0	0
1	1	1	1
1	1	1	1

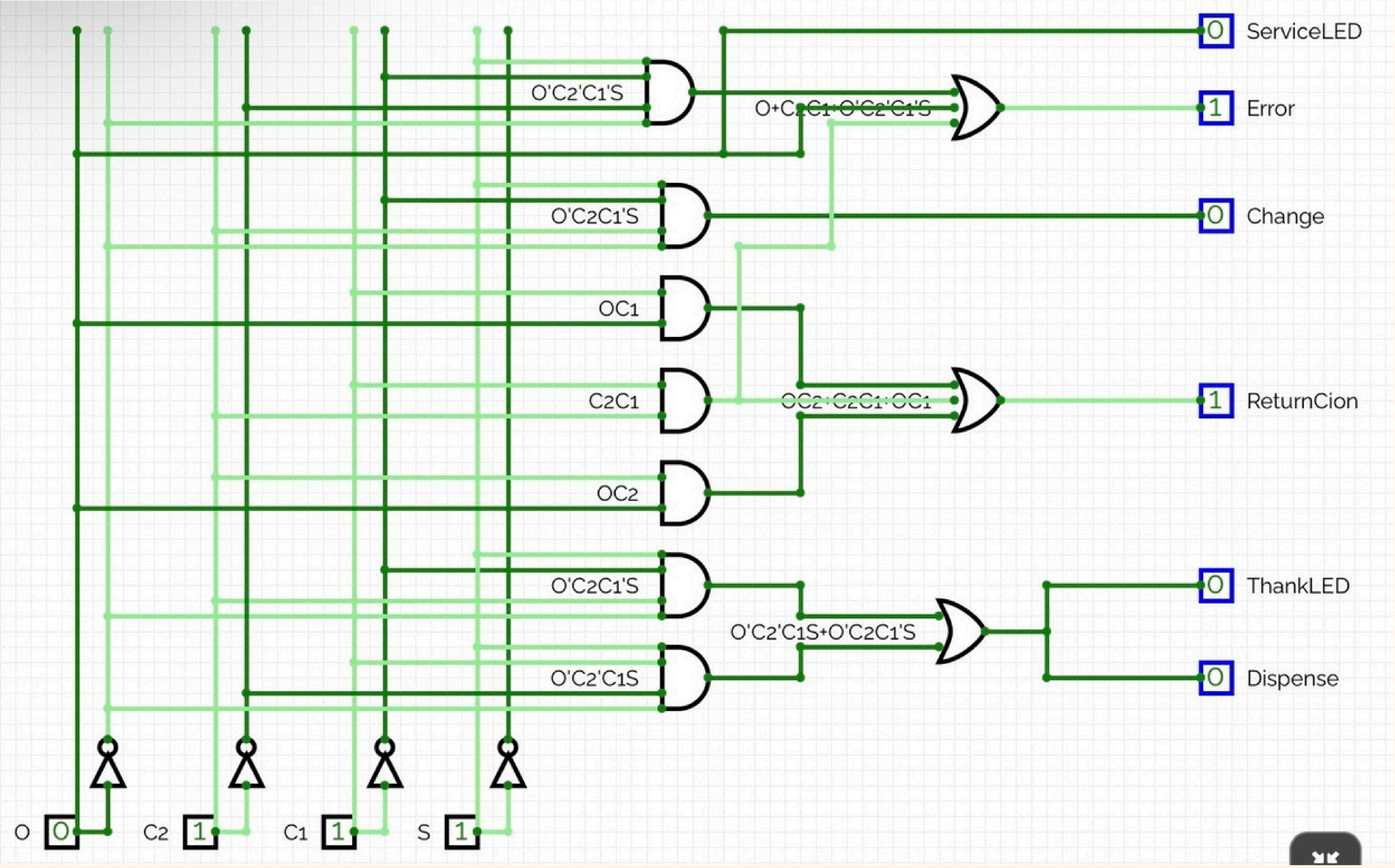
Minterms: Σ (8,9,10,11,12,13,14,15)

SOP =O





MINI VENDING CONTROLLER



ACCESS CONTROL & INDICATORS

Inputs

- A (Admin)
- G (Guest)
- D (Door)
- F (Fire)

Outputs

- Grant
- Deny
- RedLED
- GreenLED
- Lock
- Evacuate

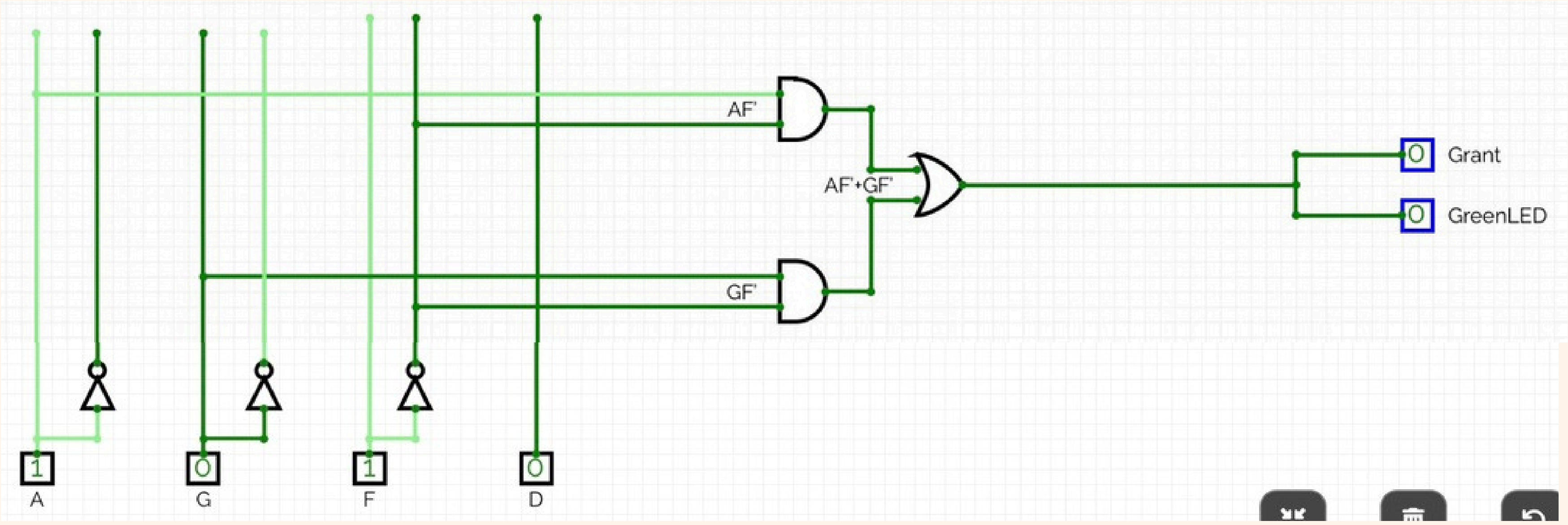
ACCESS CONTROL & INDICATORS

- Grant

0	0	1	0
1	1	0	1
1	0	0	1
1	0	0	1

Minterms: Σ (4, 6, 8, 10, 12, 14)

SOP = A F' + G F'

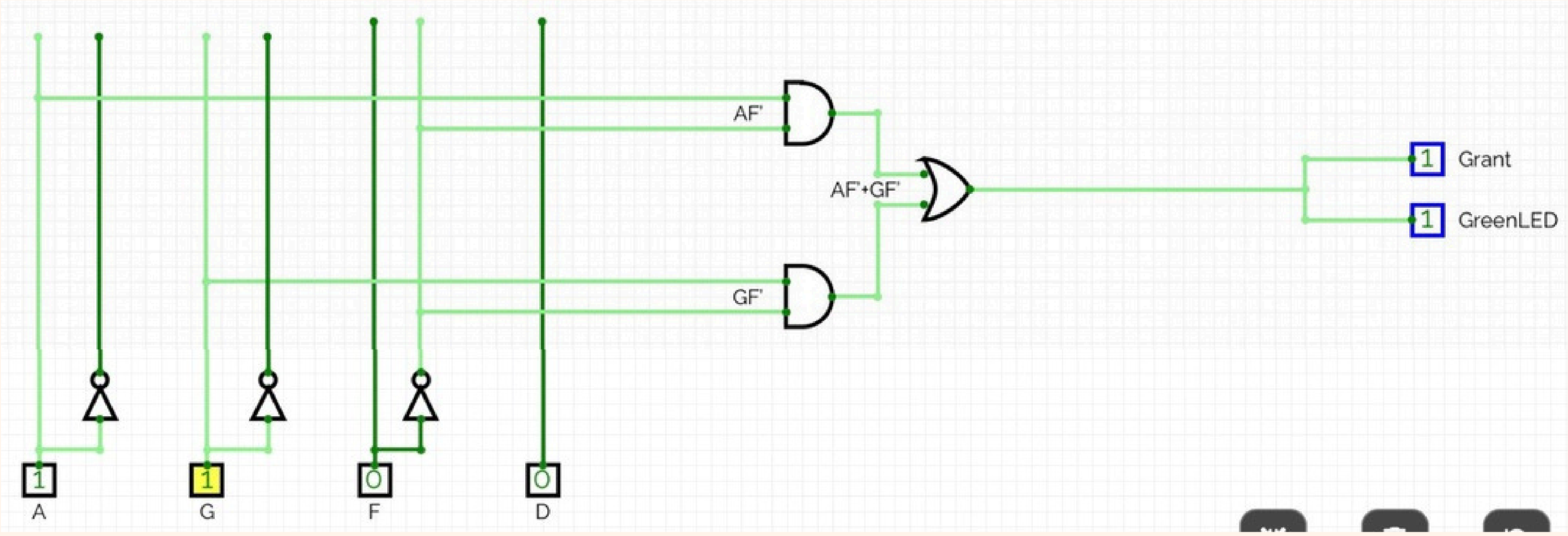


- GreenLED :

0	0	1	0
1	1	0	1
1	0	0	1
1	0	0	1

Minterms: Σ (4, 6, 8, 10, 12, 14)

SOP = A F' + G F'



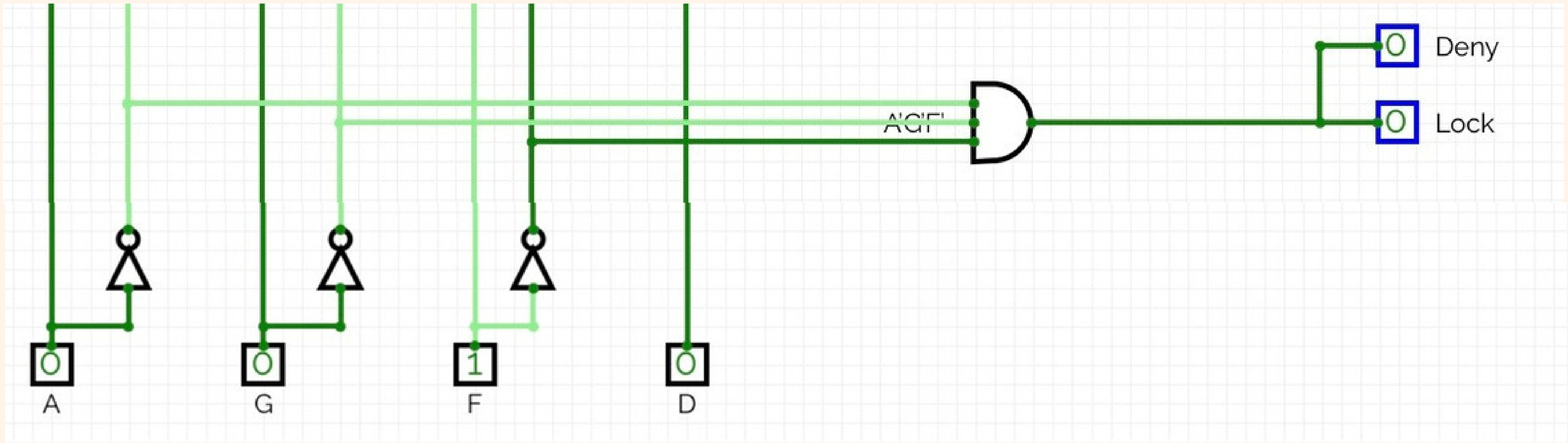
ACCESS CONTROL & INDICATORS

- Lock :

1	0	0	1
0	0	0	0
0	0	0	0
0	0	0	0

Minterms: Σ (0, 2)

SOP = A' G' F'

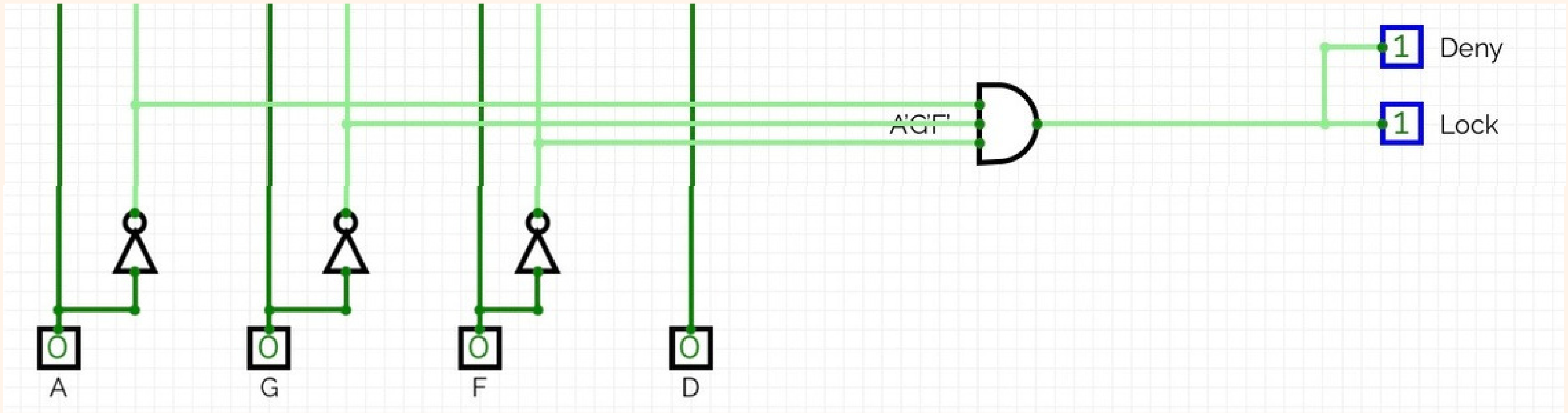


- Deny

1	0	0	1
0	0	0	0
0	0	0	0
0	0	0	0

Minterms: Σ (0, 2)

SOP = A' G' F



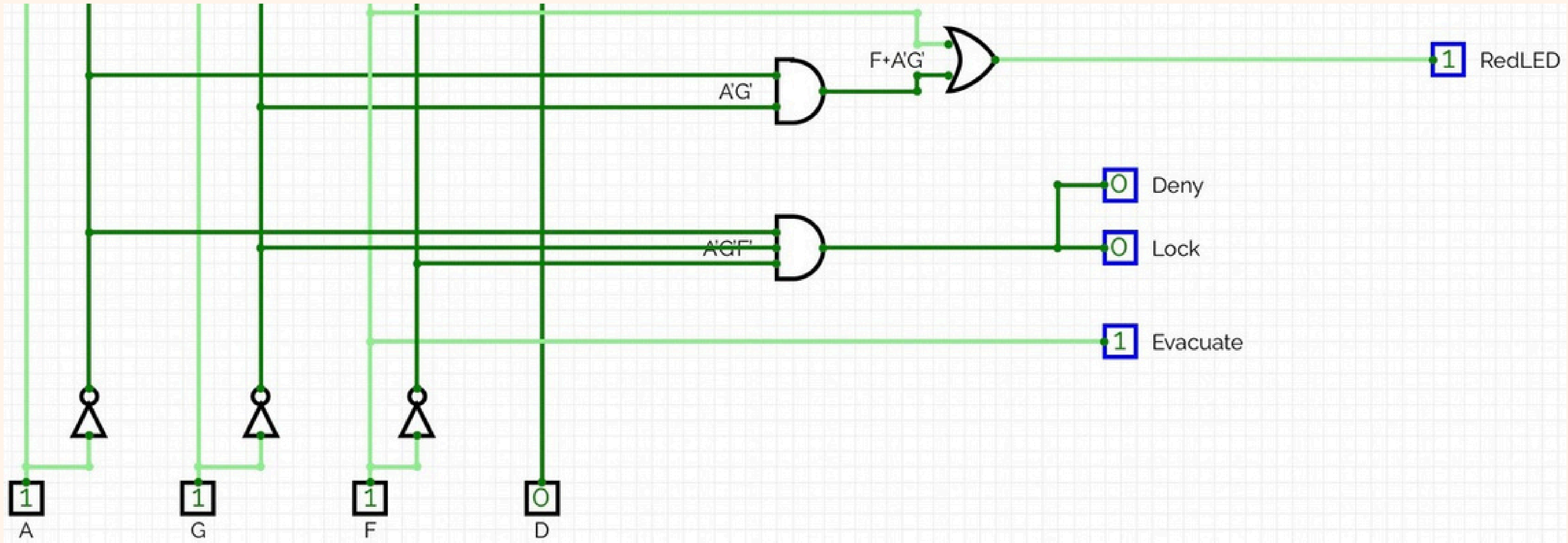
ACCESS CONTROL & INDICATORS

- RedLED :

1	1	1	1
0	1	1	0
0	1	1	0
0	1	1	0

Minterms: Σ (0, 1, 2, 3, 5, 7, 9, 11, 13, 15)

SOP = F + A' G'

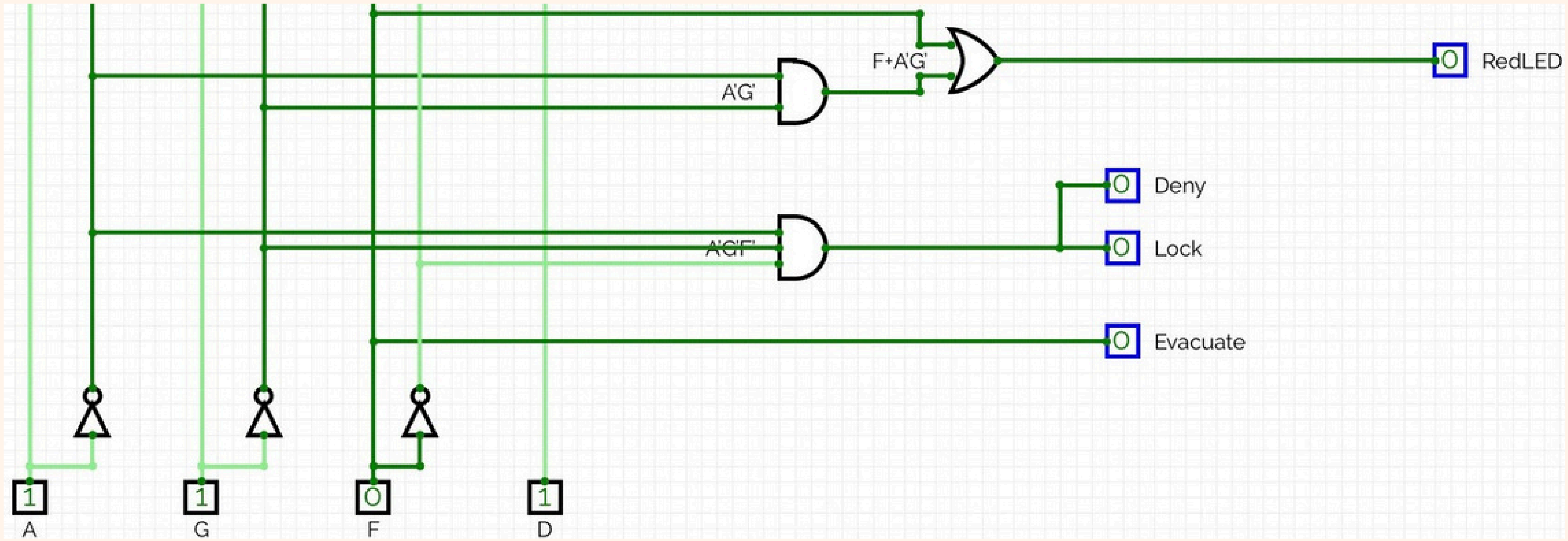


- Evacuate :

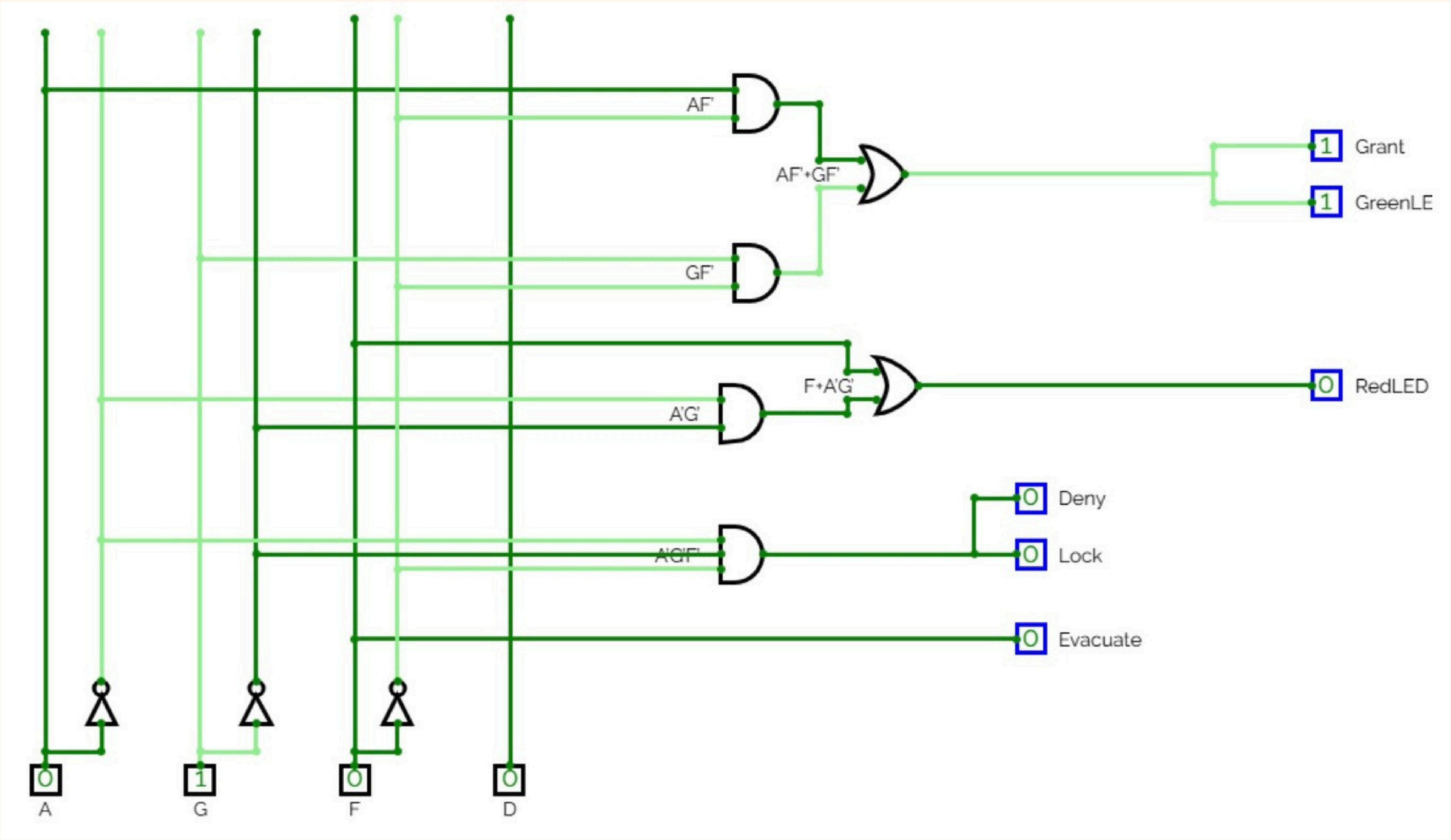
0	1	1	0
0	1	1	0
0	1	1	0
0	1	1	0

Minterms: Σ (1,3,5,7,9,11,13,15)

SOP = F



ACCESS CONTROL & INDICATORS



USAGE SUGGESTIONS

University gate

Machines in streets

THANK YOU